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An Project Report on

NLP ON MECHANIC REPAIR LOGS

Submitted in partial fulfillment for the award of degree of

Bachelor of Engineering

Submitted by

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ABSTRACT

Mechanic logs are often written in short, informal, and inconsistent language, which makes them difficult to understand for customers and sometimes even for service records. Different mechanics may describe the same issue in different ways, leading to confusion and lack of clarity. This project focuses on simplifying such logs using basic Natural Language Processing techniques.

The system takes user input describing a vehicle issue and processes the text by converting it into a cleaner and more usable form. It applies steps such as tokenization, removal of stopwords, and filtering of relevant words. The processed input is then compared with a predefined dataset containing common vehicle problems and their corresponding solutions. Based on the similarity between the input and stored data, the system identifies the most relevant issue and displays an appropriate solution.

The application is built using Python with a simple web interface, allowing users to enter their problem and select the type of vehicle. The output is presented in a structured and easy-to-understand format. This approach helps in reducing ambiguity in mechanic logs and improves communication between users and service providers. Overall, the project demonstrates how basic NLP techniques can be effectively used to handle real-world unstructured text data in a simple and practical way.

INTRODUCTION

The automotive repair industry generates a large amount of technical documentation daily, especially in the form of mechanic repair logs. These logs are usually written using technical terms, abbreviations, and brief descriptions that are easy for mechanics to understand but difficult for vehicle owners to interpret. For example, phrases like “engine misfiring during acceleration with intermittent check engine light” can be confusing for someone without a technical background.

This creates a communication gap between service technicians and vehicle owners. Customers may not fully understand the issue, the required repairs, or the associated costs, which can lead to confusion and reduced trust in the service process.

Additionally, the unstructured nature of these logs makes them difficult to analyze or maintain consistently, as the same issue can be described in different ways. Therefore, there is a need for a system that can simplify and standardize mechanic logs, making them easier to understand and more useful for both customers and service providers.

1. Problem Statement

Mechanic logs are commonly used in vehicle service centers to record issues, repairs, and maintenance activities. However, these logs are usually written in short, informal, and unstructured text. Mechanics often describe problems using abbreviations, incomplete sentences, or personal writing styles. As a result, the same vehicle issue may be described in many different ways. This lack of consistency makes it difficult for users, customers, and even service systems to clearly understand the exact problem and its solution.

Mechanical logs are often unstructured and difficult to understand, making analysis time-consuming. Customers may also struggle with technical terms, creating a communication gap with service providers. With the rise of digital systems, there is a need for automated tools to interpret these logs. Using Natural Language Processing (NLP), key information can be extracted and matched with known vehicle problems to provide simpler explanations.

Therefore, the main objective of this project is to develop a system that can take a user’s description of a vehicle issue, process the text, and match it with a dataset of common problems and solutions. The system aims to convert unstructured input into a clearer and more meaningful output, making mechanic logs easier to understand and use.

2. Literature Review

2.1 LLM-based Maintenance Analysis (ArXiv)

Year: 2025
Link: https://arxiv.org/abs/2511.18727

Problem:
Traditional ML models lack deep semantic understanding.

Methodology:

- Large Language Models (LLMs)
- Transformer-based embeddings

Result:
Better semantic understanding and prediction.

Inference:
LLMs are the future of maintenance NLP systems.

2.2 Technical Language Processing for Maintenance Work Orders

Year: 2021
Link: https://www.intechopen.com/chapters/75708

Problem:
General NLP models fail on technical maintenance text.

Methodology:

- Domain-specific NLP
- Custom preprocessing

Result:
Improved understanding of technical logs.

Inference:
Domain adaptation is necessary for mechanical datasets.

2.3 NLP Model for Managing Maintenance Requests (MDPI)

Year: 2020
Link: https://www.mdpi.com/2739964

Problem:
Manual classification of maintenance requests is inefficient.

Methodology:

- CNN + LSTM
- Text classification

Result:
~78% classification accuracy.

Inference:
Deep learning can automate repair log classification.

2.4 COLING 2020 Demo: NLP System for Event Extraction

Year: 2020
Link: https://aclanthology.org/2020.coling-demos.2.pdf

Problem:
Difficulty in extracting meaningful events from unstructured text.

Methodology:

- Event extraction
- Context-aware NLP models

Result:
Structured events extracted from raw text.

Inference:
Can be used to detect failure events in mechanic logs.

2.5 NLP-based Framework for Early Identification of Design Reliability Issues

Year: 2024
Link: https://www.sciencedirect.com/science/article/pii/S221282712400773X

Problem:
Difficulty in identifying early-stage product defects from unstructured warranty and repair data.

Methodology:

- Text mining
- NLP preprocessing
- Pattern detection from repair + customer data

Result:
Early detection of reliability issues before large-scale failures.

Inference:
NLP can significantly improve early defect prediction, useful for EV maintenance logs.

2.6 SmarTxT: NLP Approach for Vehicle Defect Investigation

Year: 2022
Link: https://journals.sagepub.com/doi/pdf/10.1177/03611981221125744

Problem:
Manual analysis of vehicle complaints and repair reports is slow and inefficient.

Methodology:

- Topic modeling (LDA)
- Clustering techniques

Result:
Automatically groups similar defect patterns.

Inference:
Clustering can help identify common failure patterns in mechanic logs.

2.7 NLP and Deep Learning for Automated Vehicle Diagnostics

Year: 2022
Link: https://www.sciencedirect.com/science/article/pii/S2666827022000998

Problem:
Unstructured service reports are hard to interpret for diagnostics.

Methodology:

- Deep learning models
- NLP feature extraction

Result:
Improved accuracy in predicting vehicle faults.

Inference:
Deep learning enhances automated diagnosis from text logs.

2.8 Automotive Fault Nowcasting with Machine Learning and NLP

Year: 2023
Link: https://link.springer.com/article/10.1007/s10994-023-06398-7

Problem:
Delay in detecting faults from large-scale service data.

Methodology:

- Transformer models (BERT)
- Large dataset processing

Result:
Real-time fault prediction with high accuracy.

Inference:
Transformer models are highly effective for large-scale repair log analysis.

2.9 Intelligent Text Correction in Automotive Documentation

Year: 2024
Link: https://iaeme.com/Home/article_id/IJCET_16_01_196

Problem:
Noisy and inconsistent technical text.

Methodology:

- NLP-based text correction
- Preprocessing techniques

Result:
Cleaner and standardized text data.

Inference:
Preprocessing is critical for accurate NLP results.

2.10 NLP for Quality Control in Manufacturing

Year: 2023
Link: https://dlbasr.us/index.php/dlbasr/article/view/15

Problem:
Unstructured inspection reports hinder defect detection.

Methodology:

- Named Entity Recognition (NER)
- Text classification

Result:
Accurate extraction of defects and components.

Inference:
NER helps in structuring mechanic repair data.

3. Solution Concept

The system is designed to take a user's description of a vehicle issue and match it with a dataset of known problems.

- The user enters a log describing the issue
- The system processes the text
- It compares the processed input with stored problems
- The best matching problem is identified
- The corresponding solution is displayed

The idea is to convert raw, unstructured input into a meaningful and helpful output.

4. Methodology

The working of the system is divided into the following steps:

1. **Input Collection**
   The user enters a vehicle issue and selects the vehicle type using a web interface.
2. **Text Preprocessing**
   - Convert text to lowercase
   - Tokenize the text into words
   - Remove stopwords (common words like "the", "is", etc.)
   - Keep only useful words
3. **Data Processing**
   - Dataset contains common problems and their solutions
   - Each problem is also preprocessed
4. **Matching Process**
   - Compare user input words with dataset words
   - Count common words (intersection)
   - Select the problem with the highest match score
5. **Output Generation**
   - Display the matched problem and solution
   - Show the selected vehicle type

5. Technologies Used

- Framework: Flask
- Programming Language: Python
- Libraries:
  - NLTK (for tokenization and stopword removal)
  - Pandas (for dataset handling)
- Frontend: HTML, Tailwind CSS
- Dataset: CSV file containing problems and solutions

6. Feasibility Analysis

Technical Feasibility : The project is easy to implement using Python and available NLP libraries. No complex models are required.

Economic Feasibility : All tools used are open-source, so the project does not require any extra cost.

Operational Feasibility : The system is simple to use. Users only need to enter a problem description and select a vehicle type.

7. System Architecture

Flow of the system:

User Input → Text Preprocessing → Dataset Comparison → Best Match Selection → Output Display

- The frontend collects input
- Flask processes the request
- NLP functions clean and analyze the text
- Matching logic finds the best result
- Output is displayed on the webpage

8. Results / Output

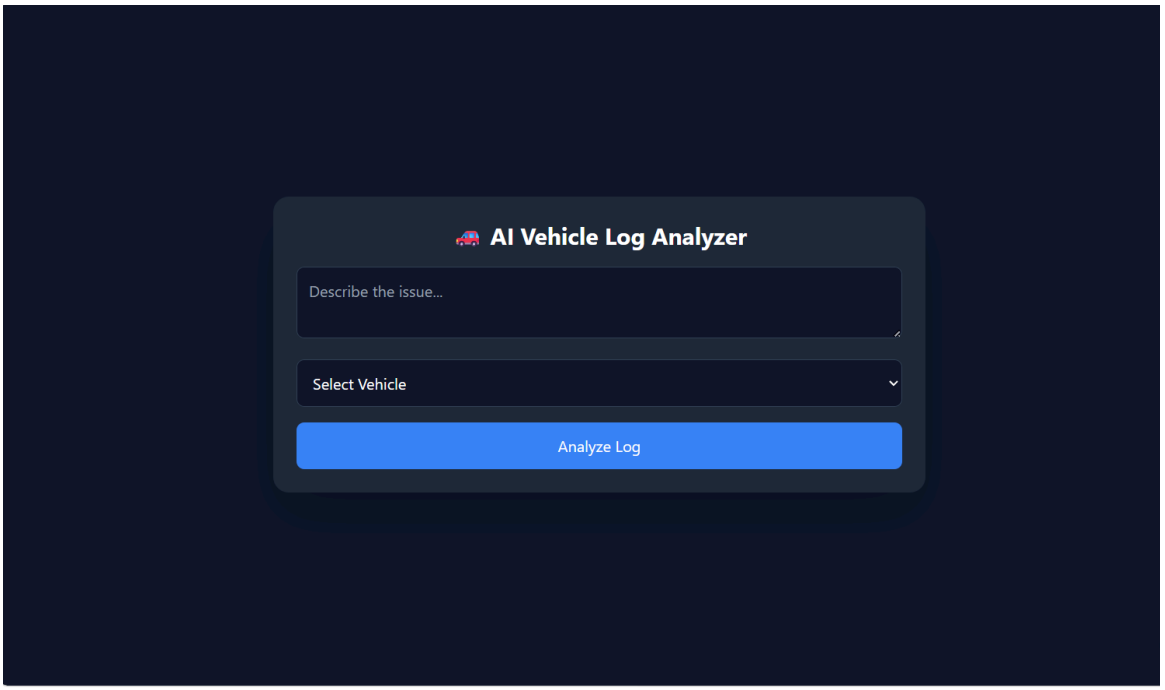

Fig 8.1: User interface of the AI Vehicle Log Analyzer for inputting and analyzing vehicle repair issues.

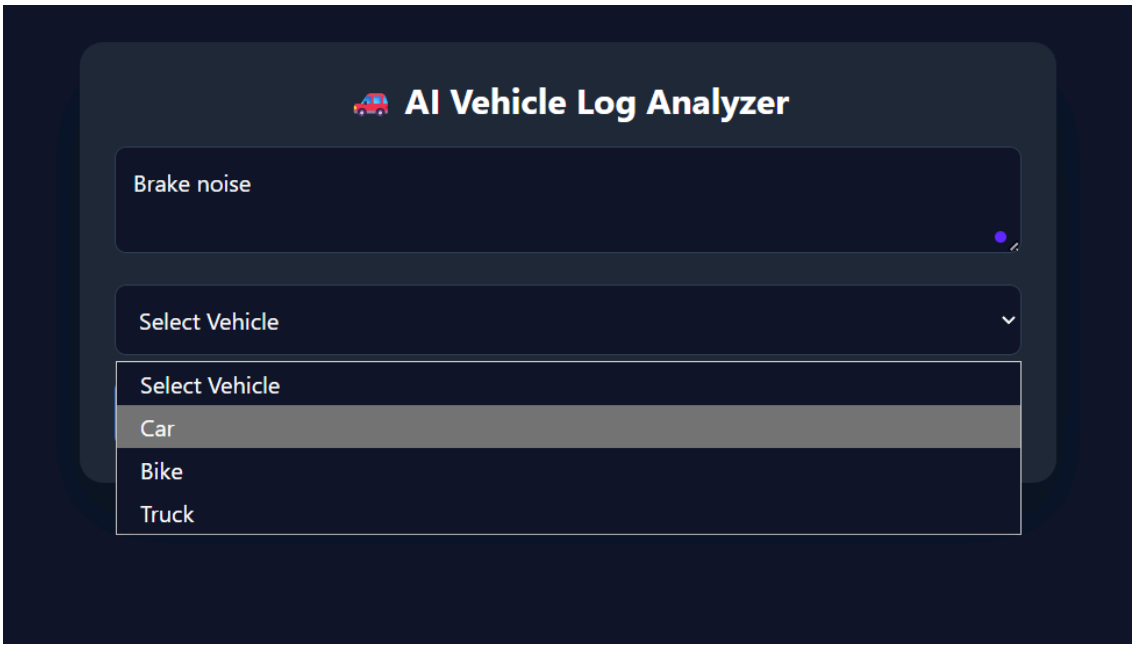

Fig 8.2 : An AI-powered interface that analyzes vehicle repair logs to identify issues based on user input and selected vehicle type.

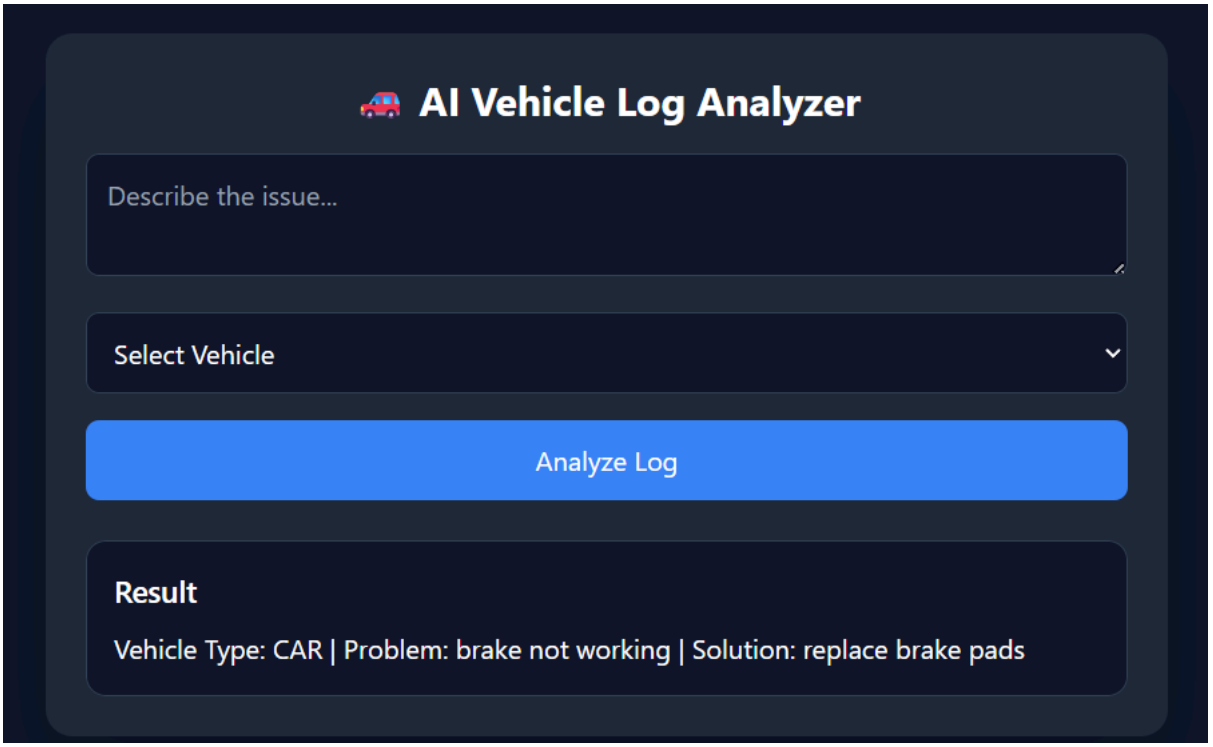

Fig 8.3 : Output screen of the AI Vehicle Log Analyzer displaying detected issue and suggested solution based on user input.

9. Advantages and Disadvantages

Advantages :

- Reduces confusion in understanding logs
- Fast processing
- Works well for common problems
- No need for complex AI models
- Simple and easy to use

Disadvantages :

- Limited to predefined problems
- May not work well for complex or long sentences
- Does not understand context deeply
- Depends on dataset quality

10. Future Scope

- Add synonym handling for better matching
- Expand dataset with more vehicle issues
- Add voice input support
- Convert into a mobile application
- Integrate with real-time vehicle diagnostic systems
- Improve accuracy using advanced NLP models